NIKHIL GOUD PALAMAKULA

Full-Stack Developer | React · TypeScript · Node.js · NestJS · MongoDB

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PROFESSIONAL SUMMARY

Full-Stack Developer with 5+ years of engineering experience, including 3+ years building production web applications on React, TypeScript, Node.js, NestJS and MongoDB. Technical lead on a multi-tenant enterprise ERP; owns features end to end from schema and API design through to UI, working directly with business stakeholders.

KEY HIGHLIGHTS

- **Co-led delivery** of an enterprise ERP as one of two technical leads in a 6-person team; owned data architecture, schema design and coding conventions across a ~89-module backend serving ~200 daily users.
- **Sole full-stack developer** on a B2B analytics platform — 45+ REST API modules and a 40+ module React frontend serving four distinct user roles.
- **Improved API response times** by ~50% (approx. 3s to 1.5s) through database indexing, query tuning and pagination, and cut recurring data-fetching cost by ~$1,100 through caching and fetch consolidation.
- **Hardened application security** with JWT authentication, refresh tokens, role-based access control and scoped service tokens across multi-tenant products.

TECHNICAL SKILLS

Frontend: React 19, Redux, React Hooks, TypeScript, JavaScript (ES6+), Material UI, Ant Design, Vite, Highcharts, HTML5, CSS3, SCSS, Responsive Web Design

Backend: Node.js, NestJS, Express.js, REST API design, JWT authentication, RBAC, WebSockets (Socket.IO), Cron jobs, Server-side PDF generation (EJS)

Databases & Cloud: MongoDB, Mongoose, SQL, Redis, Schema design, Indexing, Query optimisation, Docker, Docker Compose, AWS, Git, GitLab, CI/CD, Nginx

Practices & Domain: Agile delivery, Code review, Multi-tenant architecture, Performance optimisation, Technical leadership; PLC/SCADA, HMI, 21 CFR Part 11 compliance, Audit trails, IoT telemetry

PROFESSIONAL EXPERIENCE

Full-Stack Developer | PatternLab.ai — May 2024 – Present

Building multi-tenant B2B SaaS platforms for manufacturing, energy and export businesses.

SeaTrace — Enterprise ERP Platform — *React, TypeScript, NestJS, MongoDB* | *app.seatrace.ai*

- Co-led delivery as one of two technical leads in a 6-person team, owning schema design, data access patterns and coding conventions across a ~89-module backend serving ~200 daily users.
- Architected the exports and finance domain end to end — export transactions with server-side PDF generation, a strategy-pattern payments module and payments-outstanding tracking.

Factory Intelligence — Multi-Tenant SaaS Analytics — *React, TypeScript, NestJS, MongoDB* | *app.patternlab.ai*

- Redesigned downtime analytics across 6 client organisations so idle machine time counts toward stoppage aggregation, converting unexplained gaps into attributed, reportable capacity.
- Engineered real-time machine visibility via an authenticated Socket.IO gateway with per-tenant isolation, and cut recurring data-fetching cost by ~$1,100 through caching and fetch consolidation.

EMS — Energy Management Platform — *React, Node.js, MongoDB, Docker*

- Sole full-stack developer: designed the MongoDB schema and built 45+ REST API modules and a 40+ module React frontend serving four user roles.
- Cut API response times by ~50% (3s to 1.5s) via indexing, query tuning and pagination; secured access with JWT, refresh tokens, bcrypt hashing and role-based permissions.

Full-Stack Developer | Matrical Technologies — Jun 2023 – May 2024

- Developed a Hospital Management System covering the end-to-end patient journey from walk-in to discharge — 4 modules spanning patient records, billing, pharmacy and inventory.
- Gathered requirements directly from clients and demonstrated the platform to prospects, contributing to new business through pre-sales demos and feedback-driven product improvements.

Factory Automation Engineer | SR Automation — Nov 2020 – Jun 2023

- Delivered automation and 21 CFR Part 11 compliance projects across 4 pharmaceutical companies — Hetero, MSN, Aurobindo and Dr. Reddy's — commissioning Beijer HMI systems in place of legacy Mitsubishi and Allen-Bradley PLCs.
- Captured requirements on site with plant engineering teams, designed SCADA interfaces with SQL, and implemented user management, audit trails and electronic reporting to meet regulatory requirements.
- Programmed and commissioned PLC systems (Mitsubishi, Allen-Bradley) while owning techno-commercial delivery — client negotiations, new orders and quotation-to-closing, including Mahindra & Mahindra.

EDUCATION

MBA, JNTU Hyderabad (2025 – Present, pursuing) · **B.Tech, Electronics & Instrumentation Engineering**, JNTU Hyderabad (2016 – 2020)